

Chapter Genetics

1. A male gorilla has 48 chromosomes in each of its body cells. How many chromosomes will each of the sperms have? MAIN 2024



- (a) 24
- (b) 48
- (c) 12
- (d) 16

2. The type of gene, which in the presence of a contrasting allele is not expressed : COMP 2021

- (a) Homozygous
- (b) Heterozygous
- (c) Dominant
- (d) Recessive

3. A homozygous pea plant having purple flowers is crossed with a homozygous pea plant bearing white flowers. The phenotypic ratio of the offspring obtained in F₂ generation is : MAIN 2021

- (a) 2 : 1
- (b) 1 : 1

(c) 1: 2 : 1

(d) 3 : 1

4. The genotype of a person who cannot roll his tongue is_____.

MAIN 2021

(a) Rr

(b) RR

(c) rr

(d) RRr

5. The basic units of heredity are_____ .

SQP 2021

(a) Chromosomes

(b) Chromatids

(c) Genes

(d) Centrosome

6. ____is the scientific name of garden pea, which Mendel used for his experiment. **COMP 2023**

7. The plant used by Mendel was sweet pea. **MAIN 2023**

8. Mutation **MAIN 2022**

9. Father of Genetics. **MAIN 2023**

10. What are autosomes ? **MAIN 2023**

11. What is a phenotype ? **SQP 2021**

12. Explain the structure of a chromosome. **SQP 2023**

Solution

1. (a) 24

During meiosis, the chromosome number is halved, so each sperm of a male gorilla has **24 chromosomes**.

2. (d) Recessive

A **recessive allele** does not express itself when a **dominant allele** is present. It shows its effect only in the **homozygous condition**.

3. (d) 3 : 1

In the F_2 generation, the phenotypic ratio is **3 purple : 1 white**, as purple is **dominant** over white.

4. (c) rr

Tongue rolling is a **dominant trait**. A person who **cannot roll** their tongue has the **recessive genotype rr**.

5. (c) Genes

Genes are the **basic units of heredity** that carry the **genetic code** responsible for an organism's traits.

6. Pisum sativum

7. False. The plant used by Mendel was garden pea

8. Mutation is the sudden change in a gene which alters the hereditary materials of an organism.

9. Gregor Johann Mendel

10. The chromosomes that do not have genes for sex determination are called autosomes.

11. The observable characteristics of an organism that are genetically controlled is called phenotype.

12. During **metaphase**, chromosomes condense and replicate to form **two identical chromatids** joined at a **centromere**. Each chromosome carries **genes** that are responsible for the **transmission of hereditary traits** from one generation to the next.